

Question 1

1.1

The technical debt in case A is **entanglement**, where changing one feature can affect the functionality of other features. In case B, the debt is **undeclared consumers**, while case C represents **glue code debt**.

1.2

I selected problem A. One way to address this issue is to introduce an intermediate layer before the estimated delivery time reaches the prediction model. The purpose of this layer would be to convert the delivery time using the previous successful rounding logic that produced correct recommendations. This would isolate the change and reduce its effect on the rest of the system.

Question 2

For this question, I used the **MNIST** dataset and trained neural networks using six different experiments. The two hyperparameters varied across the experiments were the **network architecture** and the **learning rate**. The results of all six experiments were recorded using MLflow, and the corresponding screenshots are included in the IPYNB file as evidence of the experiments.

Question 3

The tasks for Question 3 were completed and the corresponding screenshots and outputs are provided in the IPYNB file as evidence of the work carried out.

Question 4

Question 4 was the most testing part of the assignment for me. I encountered several errors while setting up and connecting the different components, but working through them helped me understand the complete reproducibility workflow much better. I worked as **Partner A** for this question.

The project uses Git/GitHub for code versioning, DVC for dataset versioning, Amazon S3 for remote data storage, and MLflow for experiment tracking and model management. The repository also includes the environment configuration required to reproduce the experiment.

The complete implementation and instructions are available in the GitHub repository:

https://github.com/hari12-hp/DA3408_1_QUESTION4

Partner B independently reproduced the workflow and contributed their results as part of the collaboration.